Characterizing Reliability and Failure Regimes of Memory-Centric Neural Inference Under Coupled Precision and Hardware Non-Idealities

256×256 MRAM pathfinding: covariance-aware mapping

Du Yanzhang

Department of Electrical and Computer Engineering
National University of Singapore

Supervisor:	Prof. Fong Xuanyao Kelvin
Date:	4 November 2026
Phase:	Candidate development and bounded screening
Evidence:	Literature, software and a constructed model
Target circuit results:	Not yet available in this phase

This working draft retains the original precision–nonideality objective. Covariance-aware mapping (F3) is the next bounded intervention under investigation; Ising (F7) remains a conditional alternative. Neither physical validation nor novelty is established. The official CA2 marking rubric has not been supplied.

Contents

1 Research problem and objective **1**
1.1 Research hierarchy and proposal continuity . . . 1
1.2 Contribution filters and acceptance gates . . . 1
1.3 Evidence policy . . . 2

2 Related work and candidate gaps **2**
2.1 Mapping, activity and conversion . . . 2
2.2 Physical mapping, calibration and correction . . . 2
2.3 Reliability estimation and decision-directed computation . . . 3
2.4 Limits, alternatives and applications . . . 3
2.5 Access limits and consequences . . . 3

3 Primary intervention and retained alternatives **3**
3.1 F1: Event-aware inference-failure estimation . . . 4
3.2 F2: Spatially informed polarity . . . 4
3.3 F3: Reliability-constrained mapping and readout . . . 4
3.3.1 A narrower candidate: covariance of decoded bit-cycle errors . . . 5
3.4 F4: Decision-directed neural refinement . . . 6
3.5 F5: Analog–digital crossover . . . 6
3.6 F6: Readout identifiability and extra observables . . . 6
3.7 F7: Ising field persistence and refinement . . . 6

4 Target architecture and model contract **7**
4.1 Cost and statistical contract . . . 7

5 Bounded experiment programme **7**
5.1 First executable workload screen . . . 8
5.2 First bounded circuit set . . . 8
5.3 E03Q: finite-ADC decoder sensitivity . . . 9
5.4 Fair comparisons and stopping . . . 9

6 Progress and research-path record **10**
6.1 Repository reset and traceability . . . 10
6.2 Completed engineering verification . . . 10
6.3 E03M: a constructed covariance check, 4 November . . . 10
6.4 E03Q: finite-ADC sensitivity (constructed-model pilot) . . . 11
6.5 External task-level leads and reproduction boundary . . . 12
6.6 E09A follow-up: exact Ising controls . . . 12

6.7 How results enter CA2 12

7 Next decisions and expected contribution **12**

7.1 Assistance and ownership 13

A Fresh 256×256 Physical Array Reconstruction (E02) **13**

A.1 Target and evidence reset 13

A.2 Construction and preflight 14

A.3 Candidate methods and current limits 14

1 Research problem and objective

MRAM compute-in-memory combines nonvolatile weight storage with parallel computation. Its useful precision and efficiency depend on the circuit that converts cell states and input activity into a sensed result. A large number of operations per read can reduce conversion overhead while increasing the demands on current handling, sensing and interconnect. Existing MRAM prototypes and calibration work demonstrate why the memory device and readout architecture must be considered together [1, 2, 3].

The target of this phase is a **physical** 256×256 **array**, containing 65,536 MTJ positions. The original proposal concerns reliability and failure regimes under coupled numerical precision and hardware nonidealities. The current task is to develop a specific method or design contribution within that subject, with evidence that exceeds another parameter sweep. Earlier work is preserved in an archive and supplies no target-array validation in this draft.

The governing question is:

> Which prediction, encoding, organization or readout method can improve the cost of useful reliable computation on the declared target, and what physical mechanism makes the improvement possible beyond tuned existing methods?

The question has seven competing method families. Broad techniques such as calibration, mixed precision, activation grouping and speculative refinement already exist. Their existence determines the comparison; it does not establish that every narrower question has been answered. Conversely, using MRAM instead of another resistive technology does not itself establish originality.

1.1 Research hierarchy and proposal continuity

The new research phase has one primary *development* path, not one proven contribution. Its foundation is matched neural inference and the precision–nonideality failure-regime analysis specified in the original proposal. F3, a resident row mapping intended to control decoded cross-cycle error covariance, is the next bounded intervention to test. F7 investigates an independent Ising field-persistence question only if its physical mechanism and scope warrant continuation. F1 is a potential evaluation tool; other families remain recorded alternatives and necessary comparators.

The uploaded 4 November revision states that the original proposal's CA2 milestones include initial modelling, a baseline network, representative perturbations and preliminary inference and failure-regime criteria. These are project-planning deliverables, *not* an independently verified official marking rubric. No new supervisory endorsement is inferred.

1.2 Contribution filters and acceptance gates

Each candidate is screened for (i) useful value at fixed task quality, (ii) a distinct technical difference, (iii) physical credibility, (iv) bounded FYP scope and (v) advantage over simpler tuned alternatives after overhead. Three gates precede a large evaluation:

1. Specify the algorithm, derivation or circuit intervention and its exact difference from the closest applicable method.
2. Establish one physically justified operating case in the target architecture.
3. Demonstrate useful value in a small comparison against strong baselines, including calibration, model setup and additional control or hardware cost.

F3 is prioritized for this bounded investigation; no family has passed all three gates or been approved as the final novel contribution. An experiment can reject a defined contribution; it does not create originality automatically. A failed method remains a rejected method unless a different, explicitly justified research objective is adopted.

1.3 Evidence policy

This draft distinguishes exact software verification, workload structure, illustrative models, target circuit simulation, validated surrogates, task evaluation and measured silicon. Operation counts and activity proxies are not energy or reliability measurements. The reset completed software verification; the 4 November E03M follow-up adds a constructed mathematical/model example. A separate exploratory 256-wide digits-workload screen has been documented, but it is not the frozen FYP workload or target circuit evidence. No new validated target circuit or end-to-end physical inference result is introduced.

Academic sources are used at their actual access depth. A public abstract, inaccessible submission or reproduced patent description cannot establish the absence of an implementation feature. The literature register records these boundaries.

2 Related work and candidate gaps

2.1 Mapping, activity and conversion

VARA's CAR algorithm groups co-zero input dimensions and applies corresponding weight permutations [4]. DIB-CIM adjusts activated input blocks and ADC resolution [5]. HINT and other sparsity-aware sensing designs further constrain a generic input-aware MRAM contribution [6, 7, 8]. Weight splitting and operational parallelism have also been optimized jointly [9].

Crafton et al. model ADC-code probabilities from device statistics, profile each binary VMM and optimize a bit-significance-dependent parallelism LUT under an aggregate MAE constraint [10]. Their explicit weighted marginal approximation is a close comparison for any proposed cross-cycle error score; bit significance alone is not new.

These papers make ordinary grouping, sparsity-aware conversion and read parallelism necessary comparators. The residual F3 question is whether a *target-circuit-dependent* grouping condition and constrained mapping produce a useful advantage after the comparators are tuned. The implemented CAR-style comparator follows v1's grouping rule, not its complete training or system architecture. The latest revision and DIB-CIM's full implementation remain access gaps.

2.2 Physical mapping, calibration and correction

Physical mapping combined with ADC calibration is a close alternative [11]. Dyn-A-Map targets unreliable regions through fault-aware mapping [12]; its fault model must be distinguished from nominal analog distortion. Statistical error compensation in MRAM addresses column mismatch and wire-induced nonlinearity [13]. A candidate must therefore outperform more than an uncalibrated fixed-reference circuit. The full correction implementation is needed before claiming a faithful reproduction.

Ordinary complement reconstruction has a matching public disclosure lead [14]. Activity-based bus inversion is also established [15]. The residual F2 question concerns a cheap position-dependent electrical decision, its physical validity and implementation overhead. Original-document verification and a focused encoding prior-art comparison remain necessary.

2.3 Reliability estimation and decision-directed computation

Calibrated hardware models [1], precision analysis [16], semi-analytical fault propagation [17] and probability-region bounds [18] constrain F1. DeepVigor+'s single-bit floating-point model is not automatically the same as simultaneous correlated analog errors. Small-Noise has no public preprint [19], so its project title is an overlap warning rather than proof of method equivalence.

Zhang et al. explicitly use channel shuffling to randomize deterministic column errors when composing larger computations [1]. Thus decorrelation is already a mitigation; the F3 refinement must beat ordinary shuffling at matched placement/routing and read cost, including an equal offline search budget, rather than claim that correlations matter.

Mixed precision [20], speculative slicing and recovery [21], double-reference recompute [22] and confidence stopping [23] constrain F4. The intended gap must be a specific inexpensive analog-MRAM bound and refinement implementation. Generic interval separation or selective precision is insufficient. Claimed preprint bounds require independent checking before being used for our hardware.

2.4 Limits, alternatives and applications

Intrinsic analog precision limits are established [24]. ADC architecture models can support initial cost sensitivity [25], while digital MRAM supplies a credible alternative family [26]. F5 needs a predictive, validated crossover rule or feasible hybrid. F6 needs a circuit-specific identifiability result and an economical extra observable, rather than a restatement that analog noise limits precision.

Ising device precision and adaptive stochastic-computing precision are existing topics [27, 28]. Snowball explicitly initializes fields from row-major bit planes and updates them incrementally from column-major planes [29]; the pipelined p-computer also uses initialized, flip-updated fields [30]. These are mandatory baselines. F7 consequently requires a separate sampling-method comparison. Earlier sparsity-aware architectures [31] and expert routing [32] are context and application mechanisms. Tail reliability and selective mitigation are also existing themes [33].

2.5 Access limits and consequences

The evidence register has 33 primary-paper, institutional, author-project or disclosure records. Some methods were inspected only through selected passages; several recent implementations remain abstract-only. No cross-paper ranking of headline efficiency is made. A novelty pass requires complete inspection of the closest applicable implementation, not merely a new title, a different memory technology or a missing search result.

3 Primary intervention and retained alternatives

F3 is the leading *experimental candidate* rather than a confirmed method contribution. F1 is a possible evaluator and F7 a conditional independent Ising candidate. The remaining families are deferred, not erased.

The active inventory contains 21 stable idea IDs, consolidated into seven families. The repository candidate cards give each family a method target, physical case, closest comparators and falsifier. The following relations define the starting work; they are not claimed as new mathematics.

3.1 F1: Event-aware inference-failure estimation

Represent dominant persistent circuit effects by latent parameters z, retain remaining variation conditionally, partition important directions at actual ADC decoding transitions, and integrate tractable pieces while sampling the residual. An inference reuses the same chip realization across all its reads. The difficult contribution is a compact circuit-derived representation with useful accuracy and scaling.

For signed bit-plane coefficients a_b and read errors e_b,

$$\Delta y = \sum_b a_b e_b, \tag{1}$$

$$\mathrm{Var}(\Delta y) = \sum_b a_b^2 \,\mathrm{Var}(e_b) + 2\sum_{b<c} a_b a_c \,\mathrm{Cov}(e_b, e_c). \tag{2}$$

Marginal count-error rates do not supply these covariances or determine classification failure. E06 compares total prediction cost with calibrated strong sampling methods and compatible verification methods. Boundary explosion, setup overhead or no measurable benefit is a stopping outcome.

3.2 F2: Spatially informed polarity

For 0/1 bit planes, let $K = W^T\mathbf{1}$. Then

$$W^T x = K - W^T(\mathbf{1} - x). \tag{3}$$

The decoding identity must be adapted for XNOR or complementary-cell circuits. If a validated electrical proxy is $J(a) = a^T H a$ with symmetric H,

$$J(x) - J(\mathbf{1} - x) = (2x - \mathbf{1})^T H \mathbf{1}. \tag{4}$$

Precomputing $\lambda = H\mathbf{1}$ gives a weighted input score. Uniform weights reduce to ordinary minority inversion. The proxy's link to decoding and total cost remains to be validated. Column-count metadata alone costs $9 \times 256 = 2{,}304$ bits per binary array.

Given input activity m and known column count K_j, the possible ideal overlap-count cardinality is

$$L = 1 + \min(m, 256 - m, K_j, 256 - K_j), \tag{5}$$

which is unchanged by complementation. Lower activity therefore does not automatically reduce the information that must be resolved. E04 compares calibrated direct reading, minority inversion, the electrical selector and an offline oracle, including control cost.

3.3 F3: Reliability-constrained mapping and readout

A row permutation P applied consistently to x and W preserves ideal output:

$$(PW)^T(Px) = W^T x. \tag{6}$$

Its electrical effect need not be invariant. For fixed total input activity M, R rows per group and C nonempty groups,

$$\overline{m}_{\text{executed}} = M/C, \qquad \lceil M/R \rceil \le C \le \min(M, 256/R). \tag{7}$$

The lower bound permits an input-specific packing oracle, not a universally achievable static permutation. Increasing mean executed-read activity is a concentration effect; it does not prove aggregate reliability degradation.

For a simplified expected count-error cost $c(m)$ with $c(0) = 0$, combining loads a, b changes the cost by

$$\Delta c = c(a + b) - c(a) - c(b). \tag{8}$$

Linear c leaves the expected total unchanged; a convex cost can penalize concentration. Real cost depends additionally on pattern, programmed weights, position, calibration and timing. Task failure further depends on joint errors and signed reconstruction.

An illustrative scalarized read cost $F(m) = E_0\mathbf{1}_{m>0} + E_1 m + \lambda c(m)$ yields

$$\mathbb{E}[\Delta F] = -E_0 \Pr(a > 0, b > 0) + \lambda \mathbb{E}[c(a + b) - c(a) - c(b)]. \tag{9}$$

This separates saved fixed read overhead from a concentration penalty. It is a modelling framework, not a measured MRAM energy law or an originality claim.

The candidate mapper starts from CAR and balanced orders, scores complete calibration traces using a validated response/cost model, and accepts bounded row swaps satisfying reconstructed-error, timing and area budgets. Each baseline may retune its supported readout. A generic constrained optimizer alone is insufficient; the deliverable must include a useful circuit-dependent grouping rule and a demonstrated advantage.

3.3.1 A narrower candidate: covariance of decoded bit-cycle errors

On 4 November the F3 method question was sharpened: can two mappings with equal read counts, activities and marginal decoded-error laws have different reconstructed errors after scalar calibration? For reconstruction coefficients b and decoded error vector e, the proposed score is

$$R(P) = \mathbb{E}[(b^T e)^2] = (b^T \mu_e)^2 + b^T \Sigma_e b. \tag{10}$$

Pair moments suffice for this MSE identity; they do not determine the full error PMF or classification-failure probability. A validated conditional current model and actual decoder boundaries would supply the moments. Shared periphery must remain in the model.

The score is a *search surrogate*. A frozen held-out evaluation must separately report reconstructed arithmetic error, prediction mismatch relative to the clean quantized network, and ground-truth classification error with paired input/chip uncertainty. A variance advantage alone does not establish neural reliability. If a column-correlated estimator is used to rank maps, its omitted joint events and calibration must be audited against direct persistent-chip sampling.

A single fixed random permutation and the best of random candidates found with the same offline search budget are *different* baselines. A channel-shuffling comparator is mandatory, because ordinary decorrelation is already known. Charge legal placement, routing, capacity, offline optimization, and calibration fairly.

The prospective offline mapper searches bounded row exchanges from CAR and balanced initializations, updating the affected read moments subject to a fixed resource budget. One mapping serves every activation plane on the resident array. The general search implementation remains future work; a two-map score and constructed check are complete. The distinction from CAR's co-zero affinity [4], bit-aware adaptive blocks [5], fault-region mapping [12] and marginal-code/bit-significance optimization [10] is the joint decoded-error score and its grouping consequence. This is a candidate difference, not a certified absence from prior work. Full DIB-CIM access is still required.

3.4 F4: Decision-directed neural refinement

Begin at the final classifier. A coarse computation supplies score intervals $[\underline{z}_j, \overline{z}_j]$. Accept class c when

$$\underline{z}_c > \max_{j \neq c} \overline{z}_j; \tag{11}$$

otherwise refine under a frozen policy. The interval must include omitted terms and declared hardware error. A statistical envelope requires error control valid under adaptive refinement. F4 must demonstrate an inexpensive bound and implementation; bound construction, fallback and the layer's share of total task cost are included.

3.5 F5: Analog–digital crossover

Derive the feasible region in which analog cost $E_A(m, R, B, t, z)$ is below digital cost $E_D(m, z)$ at matched precision, task quality, throughput and capacity. Here z denotes a declared calibration/chip class. A small mode-selection table is meaningful only if both modes and their extra periphery are physically supported. E07 compares best fixed analog, digital and static-hybrid alternatives. A sweep alone is insufficient.

3.6 F6: Readout identifiability and extra observables

For one observable v, economical readout metadata, class priors π_k and calibrated densities $f_k(v)$, the Bayes error floor is

$$P_e^* = 1 - \int \max_k \{\pi_k f_k(v)\}\, dv. \tag{12}$$

This is established decision theory. The intended work is a target-specific separation condition and a minimal additional sample/reference/read configuration that improves the practical trade-off. Giving the controller the true result, or evaluating the whole dot product digitally for free, would invalidate that argument. E08 checks identifiability, noise amplification and calibration alternatives before adding a sensing intervention.

3.7 F7: Ising field persistence and refinement

F7a is a conditional diagnostic: distinguish fixed manufacturing variation, physical temporal read noise, an intentionally decorrelated control and error accumulated in cached fields. Matching a mean read-error rate does not match conditional error laws. Full analog field recomputation and incremental reads of binary couplings are different physical operators. A fixed matrix and consistent initialization make exact increments telescope; fresh coefficient-read errors can leave a cache residual after a spin flips and flips back. E09B must distinguish this effect from a temperature-floor hypothesis. Generic coupling sensitivity and incremental updating are already established [27, 29, 30]. A companion chapter remains conditional on a useful specific question, shared mechanism/method and scope fit; the same noise table alone is insufficient.

Stable binary MRAM can store couplings while randomness is generated separately. For a monotone update probability $p(h)$, draw one uniform u and obtain a valid field interval $[L, U]$. Commit the positive update if $u < p(L)$, the negative update if $u \geq p(U)$, and otherwise refine while retaining u. With valid deterministic bounds, the same draw produces the exact-field decision. Noise, update order, temperature schedule and fallback must be included. E09 remains a conditional application/companion; refinement's mathematical comparison and interval hardware remain unresolved.

4 Target architecture and model contract

The target capacity is fixed, while read parallelism, scheduling and isolation are unresolved. A logical batch is a schedule within the physical array; it does not remove wire capacitance or create independent short bitlines. Physical sectors require explicit isolation devices and local/global sensing choices, with resistance, capacitance, area and timing overhead.

Item	Current declaration
Physical cell positions	256×256; binary plane mapping declared separately from logical weights.
Cell/access model	Authorized access-transistor plus student-owned MTJ source policy; target interface needs inspection.
Wire and driver model	References, topology, loading and parameter ranges unresolved.
Readout/ADC	Circuit/interface, full scale, precision, sharing and sample time unresolved.
Variation	Device identity, spatial/common components and repeat-read noise must be specified.
Calibration	Independent calibration patterns/population and tuning budget required.
Maximum active rows	Not inherited from prior trials; establish in the target model.
Validation evidence	Held-out target patterns and extraction/decoding parity required.

Table 1: Prospective architecture contract. Unknown entries are not filled with assumed measured values.

Initial parameterized models may use defensible ranges and reduced equivalents with explicit limitations. Quantitative final claims need validated models and, where appropriate, extracted parasitics. Neighboring-column and shared-driver effects must either be explicit or justified by an equivalent loading representation.

The source boundary permits authorized access-device use in the university environment with the student-owned MTJ model; it does not authorize redistributing licensed models. No proprietary foundry MTJ implementation is bundled. Actual Spectre execution and source validation take place in the authorized environment.

4.1 Cost and statistical contract

Total cost includes array and driver excitation, sensing/ADC, digital accumulation, control/data movement and leakage. Repeated conversions, rejected speculation, permutation addressing, metadata, fallback and calibration amortization are included. Useful output precision and throughput must match across analog and digital alternatives.

Mappings and settings are fitted on calibration data, then frozen. Previously inspected datasets remain development data. A device mismatch assignment persists across reads; repeated reads are not independent chips. Report ground-truth classification error, clean-reference mismatch and arithmetic error separately. Paired policies share input and chip realizations, and their uncertainty must respect that pairing.

5 Bounded experiment programme

The immediate sequence is E00 engineering verification, E01 real-trace structure and E02 target physics. Their outcome and full closest-work inspection select one main intervention and one backup. The programme does not launch all families as large campaigns. Quality, latency and

area budgets are frozen before each scientific comparison; no unjustified numerical acceptance target is introduced here.

ID	Question / first comparison	Continue or stop criterion
E00	Exact identities and metadata guards. Implemented software checks.	Arithmetic parity, disjoint IDs and non-overwrite behaviour must pass. No physics claim follows.
E01	Real-trace occupancy under original, random, zero-rate, balanced, CAR and quiet-balanced orders.	Continue F3 only for a nontrivial structural opportunity after inactive-feature controls.
E02	Target-array connectivity, pattern/location dependence, calibration and settling.	Validate a physically meaningful response model on separate patterns before reliability transfer.
E03	Grouping concentration and constrained mapper against retuned CAR, balancing, correction and adaptive blocks.	Stop for no calibrated conflict or no gain beyond a simpler tuned baseline.
E04	Direct, minority and electrical polarity versus an offline oracle.	Stop for little oracle benefit, weak score prediction or controller overhead exceeding gain.
E05	Interval-directed classifier refinement versus fixed/mixed precision and detector recompute.	Stop for invalid/loose bounds, universal refinement or no net resource saving.
E06	Event-aware failure estimator versus calibrated strong sampling / compatible bounds.	Require prediction accuracy or confidence at lower total cost, including setup and reuse.
E07	Predictive analog/digital crossover with a feasible hybrid.	Stop if one simple mode dominates or duplicated periphery removes gain.
E08	Best calibrated observable versus extra resolution or one extra measurement.	Require relevant identifiability improvement beyond simpler correction at matched cost.
E09	Exact references, persistence/cache diagnostics, then a small exact-target Ising pilot. Refinement is a reserve.	Conditional branch/companion; require a specific useful result beyond known coupling errors and incremental fields.

5.1 First executable workload screen

E01 accepts a real trace bundle with binary matrices X_{cal}, X_{eval} of width 256, one 256×256 weight plane and disjoint stable IDs. Model/quantization hashes, the layer, exporter command and data role are recorded. Padding and unused columns are explicit. Exact matrix output is checked for every mapping. One resident row order must serve all activation planes and inputs sharing the programmed array; pooled calibration traces fit that order. Per-plane optima are diagnostic oracles unless their reprogramming or routing costs are included.

The comparator set uses the calibration traces only. It reports occupancy histograms, nonempty vector-read counts, total active rows and conditional fan-in at prospective group sizes 16, 32, 64, 128 and 256. A squared-activity statistic is diagnostic; no physical error or energy law is assigned to it. Hardware and inference metrics remain unavailable. The largest skip count is not declared a winning research method.

5.2 First bounded circuit set

E02 first fixes one topology and nominal operating point. After checking current/bias limits, nominate two supported parallelism settings and 12–24 nominal pattern cases spanning zero, sparse, half-active and dense masks, equal-count physical placements and contrasting stored states. The range is a planning budget, not an executed case count. Check source extraction, KCL, driver headroom, current versus count and settling. Use additional independent patterns to validate a

reduced model, especially near decoding edges. Introduce persistent and repeat-read variation in separate arms afterwards.

The preflight currently reports missing model fields. No unverified transistor port ordering, target netlist generator or new Spectre execution is claimed by this reset. The target model must be connected to actual authorized circuit sources before running.

5.3 E03Q: finite-ADC decoder sensitivity

Before claiming a covariance-aware mapping advantage, specify the entire sensed-current transfer: input quantity and units, ADC full-scale and endpoints, bit depth, midrise or midtread code convention, reference calibration, saturation and clipping, and ties. Use nearest-*quantized*-reference decoding rather than treating an ADC name as a complete electrical specification. Compare ideal midpoint decoding, fixed finite conversion and equally budgeted reference/full-scale tuning with persistent chip identities paired across both mappings.

The constructed E03M fixture permits a diagnostic software-only E03Q extension while the actual E02 circuit remains unresolved. Its nominal active read has 16 selected cells with eight low- and eight high-current MTJs; current units are dimensionless. The pilot assumes a midrise ADC over [16,32], 4–7 output bits, and offsets $-0.25, 0, +0.25$ LSB. Calibration selects a phase using independent chip assignments; the evaluation set is held out. These settings are explicitly *illustrative*, not GF 22FDX reference, ADC or timing specifications.

Test matched read counts, marginal error laws, decoded pair covariance and reconstructed MSE; separately report signed-error tails. If marginal laws are not demonstrably matched, the MSE gap cannot be attributed solely to covariance. Full task evaluation must add fixed-random, equal-budget best-of-random, shuffling, CAR, balancing, adaptive-block and calibrated sensing baselines. A reduced MSE is a first screening result, not an inference-failure or hardware-cost claim.

5.4 Fair comparisons and stopping

The revision-15 follow-up divides Ising screening into E09A exact software controls, E09B conditional-error and cache-drift checks, and E09C a bounded solver pilot after a validated target adapter. E09A is complete; E09B/C are unrun. The first proposed pilot budget is three small graphs with enumerated optima, three independent chip assignments and twenty RNG restarts per graph/chip. These are effort limits, not a powered result. Temporal noise needs separate target provenance. A redraw of manufacturing variation at each read is a diagnostic counterfactual and is not automatically physical read noise.

Digital incremental updates are a mandatory baseline. Include initialization, coupling fetches, any transposed/duplicated storage, cache precision and refresh. Match full conditional error laws on independent fixed states before interpreting adaptive trajectories. Score the true target Hamiltonian and keep graph/chip/restart units separate. Without hardware timing, report operation budgets rather than measured TTS.

Each baseline receives the same calibration/tuning budget and may adjust supported references, ADC settings and sample time. Mark adaptations honestly where full circuit details are inaccessible. Preserve chips and input assignments across paired policies. Count correction and repeated-read overhead. An illustrative model can identify a mechanism under assumptions, but cannot substitute for target validation.

A failed candidate is retained in the decision record with its reason. A novel useful analytical rule may independently qualify as a contribution; a baseline winning a plot does not automatically become such a rule. Expand evaluation after a strong small comparison passes.

6 Progress and research-path record

6.1 Repository reset and traceability

On 3 November 2026 the user requested a new research phase and report. The complete former main tree is preserved under `archive/2026-11-03-pre-reset/`, with a named archive branch at the original commit. The manifest also records the heads of 46 pre-existing branches; their unmerged contents remain separate and traceable. Historical scope documents are not forward instructions for this phase.

The active tree records 21 idea IDs, seven families, experiments E00–E09, a claim ledger, dated logbook, decision records and a CA2 evidence map. These distinguish what was known before an experiment from what was learned after it. No narrative is reconstructed as though an untested later idea motivated an earlier measurement.

6.2 Completed engineering verification

The reset's 10 unit tests passed locally. They check exact permutation and partial-sum equivalence, signed reconstruction including the negative sign bit, complement identities, count-support cardinality, deterministic comparator behaviour, split/metadata rejection and non-overwriting records. The explicit E00 result and test transcript are curated under `experiments/results/`.

Evidence	Current status / permissible interpretation
E00 arithmetic and metadata tests	Passed software checks; no physical or task-level conclusion.
E01 real workload screen	Exploratory independent digits256 screen recorded; frozen FYP model traces remain pending.
E02 target circuit/model	Contract/preflight and protocol prepared; physical inputs and validation missing.
E03–E08 target intervention comparisons	E03Q constructed ADC sensitivity run; no target physical intervention performed.
E09 Ising alternative	Held as a separate optional application, prerequisites unresolved.
Novelty / useful hardware improvement	Not established for any candidate.

Table 3: Status at the reset. Planned work is not reported as experimental success.

6.3 E03M: a constructed covariance check, 4 November

E03M is a normalized probability model with 256×256 binary storage bookkeeping, four prospective 64-row groups and two input bit planes. It is not a target circuit simulation, physical fan-in validation, real-workload demonstration or trained mapper. The fixture uses declared current means 1 and 2 and independent 3% Gaussian current variation frozen across reads. No retired trial or foundry parameter is used.

Both mappings execute eight vector reads with 16 active rows and the same ideal count in every column. Each receives the true full-group current deviation for scalar calibration; midpoint count thresholds are ideal. Different shared-cell overlap gives analog correlation patterns $[1, -1/3, 1, -1/3]$ and $[1/3, 1/3, 1/3, 1/3]$ after Gaussian conditional-mean correction.

For decoded-error variance v and covariance $c(\rho)$, the resulting MSEs are

$$R_A = 28v - 8c(1/3), \qquad R_B = 20v + 16c(1/3). \tag{13}$$

Their difference is positive for the constructed non-linear quantizer. With continuous readout, $c(\rho) = v\rho$ and it vanishes. The mathematical expansion and proof, code, configuration and original result hashes are retained in the research record.

Constructed-model quantity	Mapping A	Mapping B
Executed vector reads	8	8
Marginal count-error probability (%)	0.234308	0.234308
Analytic reconstructed MSE	0.065222	0.047631
Continuous-readout MSE	0.684	0.684

Table 4: E03M, normalized arithmetic units squared. The 26.97% analytic MSE difference is an illustrative-model result, not a target-array improvement.

Analytic identities were checked across all 256 columns. Paired Monte Carlo on one predeclared column used three predeclared seeds with 131,072 independent variation assignments each. Conditional-mean correction produced MSE reductions of 24.47–27.88%; separate full-group gain normalization retained 24.95–27.99%. All paired difference intervals exclude zero using per-assignment standard errors. This single constructed setting establishes no advantage over a tuned CAR/adaptive-block frontier or arbitrary pattern-specific calibration, and supplies no inference or energy result.

The result motivates a bounded prior-art and target pair-covariance check under ADR-0002. It does not complete E03 or change the open final-topic decision. The curated record is `experiments/results/E03M_20261104_covariance.json`; the method and proof are in `research/novelty/2026-11-04-covariance-mapping.md`.

6.4 E03Q: finite-ADC sensitivity (constructed-model pilot)

A new E03Q software experiment extends the separate mathematical E03M fixture without using any retired small-array measurements or foundry parameters. A 256×256 binary-weight *bookkeeping* fixture retains four prospective 64-row groups, two input bit cycles and one predeclared output column. Each read has 16 selected rows with eight MTJs of each nominal current state. Manufacturing variation is independent Gaussian with assumed 3% cell-current coefficient of variation and persists across both cycles. The same 131,072 independent evaluation-chip assignments are paired across mappings; 65,536 separate chips select ADC-offset settings. A shared, oracle-like full-group mismatch correction is inherited from E03M and is not asserted physically feasible.

The finite ADC is a declared uniform *midrise* quantizer on [16,32] normalized current units followed by nearest quantized nominal references, lower-count tie-break and clipping. Four through seven bits and phase offsets of -0.25, 0 and $+0.25$ LSB are examined. The offset selected on calibration chips is frozen before the independent evaluation.

Constructed decoder	Mapping A MSE	Mapping B MSE	Relative decrease (%)
Ideal midpoint	0.068687	0.048462	29.4
Fixed 6-bit ADC	2.261963	1.917000	15.3
Tuned 6-bit ADC	0.122276	0.087631	28.3
Tuned 7-bit ADC	0.081520	0.057892	29.0

Table 5: E03Q held-out, one-column Monte Carlo. Normalized integer-output MSE; no circuit, inference or energy result.

The paired difference (A minus B) for the tuned 6-bit setting is 0.03465, with an approximate

chip-level 95% interval [0.02906,0.04023] under independent chip sampling. The fixed 4- and 5-bit operating points exhibit strong threshold-alignment artefacts, illustrating how calibration can dominate the comparison. The mappings have identical marginal Gaussian current laws by construction; their finite-sample empirical read-error frequencies are not exactly identical. This is a useful decoder-sensitivity check, but not a new MRAM mechanism or evidence of a useful trained mapper. All model inputs and output hashes are registered with the E03Q pilot.

6.5 External task-level leads and reproduction boundary

The latest externally supplied draft mentions neural mapping and noisy-Ising results from another discussion. Their raw outputs and source implementations were not independently reproduced here. They remain unverified external leads; no claimed agreement in outcome establishes a shared mechanism. Likewise, an incremental cached-field Ising error is not the same operator as full analog field recomputation.

6.6 E09A follow-up: exact Ising controls

The revision-15 consolidation added six meaningful software tests; the combined suite has 16 passing tests. E09A verifies small-graph Max-Cut optima, the target-energy/cut identity, full versus incremental sequential heat-bath trajectories under common random inputs, and exact cache parity after every update. Eight graph/seed cases include a 64-spin arithmetic-only fixture. No noisy-target success-rate benchmark was executed.

For one coupling coefficient, flipping a spin and reversing it leaves an incremental read-error residual $2s_k^{\mathrm{old}}(\eta_2 - \eta_1)$. Identical frozen coefficient errors cancel; independent zero-mean errors of variance σ^2 give residual variance $8\sigma^2$. An exact unit-sign enumeration checks this algebra. It is a cache-control diagnostic, not a measured noise law, new general theorem or target solver result. The target-kernel preflight remains blocked by missing response and temporal-noise evidence, and the scientific adapter is not implemented. Curated evidence is under `experiments/results/E09A_20261104_reference.json`.

6.7 How results enter CA2

Each meaningful run records its hypothesis, code revision, command, source hashes, calibration/evaluation roles, chip identities, evidence class, outputs, uncertainty and decision. The experiment register tracks readiness and blocking evidence. A dated decision updates the claim ledger and relevant report section. Raw runs are never overwritten; curated evidence links the analysis to the original outputs.

7 Next decisions and expected contribution

The first input task is to export one representative layer's frozen calibration and evaluation traces with their model/quantization identities. E01 then tests whether any static mapping opportunity remains beyond inactive-feature packing and simple balancing. In parallel, define the authorized target-array circuit interface and the smallest E02 connectivity, location and settling comparison. Obtain the closest full implementations before a novelty claim, especially adaptive blocking and MRAM error compensation.

F3 is prioritized for a bounded development check rather than accepted as a novel final method. Preserve the original precision–nonideality objective as the foundation. F1 remains an optional evaluator and F7 a conditional Ising experiment. The E03Q constructed finite-ADC screen is

complete; the target readout/ADC contract and physically valid 256-by-256 array E02 remain unverified. The next scientific gate is therefore new target circuit validation, not a large accuracy-versus-variation sweep.

The candidate choice remains open. F1 would contribute a demonstrably better prediction method; F2 a cheap electrically justified encoding/controller; F3 a grouping rule and effective mapping; F4 a useful bound/refinement implementation; F5 a predictive crossover rule or feasible hybrid; F6 a target-specific identifiability condition and economical sensing intervention. F7 is a separate stochastic-computing option. No candidate is selected because its title sounds newer or because its first toy model produces a loss.

Supervisor discussion should settle the permitted architecture changes, available device/wire/readout models, intended contribution type and manageable workload. The new directions are student research proposals; no new supervisory endorsement is implied. The original precision–nonideality motivation is retained while the intervention is specified and tested.

ADR-0002 gives the specific F3 covariance mechanism development priority for a bounded check. Verify decoded pair correlations for matched-activity masks on the target after equal calibration, then compare a single fitted mapping against combined tuned baselines and a covariance-ablated score. The constructed E03M result supplies a discriminating example, not the physical or practical pass.

ADR-0003 consolidates the revision-15 Ising proposal into a conditional diagnostic and possible alternative/companion. Follow the queue in `research/04_screening_queue.md`: real workload export, the target E02 gate, then small mapping and persistence/operator checks. A common story does not commit two large campaigns. Select the contribution after the strong comparisons; retain explicit negative findings and eliminated claims.

An acceptable method outcome must demonstrate either better useful reliability at a comparable resource budget, lower total cost at matched quality, or a distinct validated analytical/design rule that changes a practical decision. If the strongest simpler baseline removes the advantage, the method claim is rejected. That result and its reason remain available for honest progress reporting and subsequent topic selection.

7.1 Assistance and ownership

AI-assisted work can support literature retrieval, derivation checking, code, experiment organization, analysis and drafting. The student verifies sources, model interfaces, physical assumptions and conclusions, and runs licensed tools in the authorized environment. The maintained record distinguishes generated suggestions from executed experiments and independently reviewed claims. Actual hardware evidence and originality cannot be supplied by a narrative alone.

A Fresh 256×256 Physical Array Reconstruction (E02)

A.1 Target and evidence reset

The target is a physical 256×256 STT-MRAM compute-in-memory array with 65,536 binary-MTJ positions. A 256-by-256 software weight matrix is not a physically implemented array. All new target electrical read-current distributions, sensing-reference calibration, decoding models and layout/parasitic validation must be generated afresh. Retired small-cell and eight-cell readout trials are historical records only and do not establish target-array performance.

A.2 Construction and preflight

1. Audit the authorized GF 22FDX MTJ and access-transistor models, ports, safe read/write bias and design rules; justify the selected 1T1MTJ or 2T1MTJ implementation.
2. Construct repeatable physical tiles with verified DRC/LVS and representative BL, SL and WL parasitic extraction.
3. Define the 256-row/256-column hierarchy: distributed wire lengths and loads, selected/unselected devices, read drivers, ADC/reference, column sharing and read parallelism.
4. Perform *new* KCL, headroom, current and settling preflights using sparse, medium, dense and spatial read masks before fitting statistical error laws.
5. Verify a model against independent held-out target-circuit patterns and ADC decisions near thresholds, preserving fixed device mismatch within virtual chips and keeping temporal noise separate.

Physical isolation of subarrays must be implemented explicitly; a logical row grouping does not shorten a bitline. A hierarchical, PEX-informed circuit model must not be described as a fully extracted 256-by-256 macro.

A.3 Candidate methods and current limits

E01 may evaluate real neural activation traces as software workload evidence without target circuit parameters. The F3/E03 covariance-aware mapping hypothesis requires newly validated *decoded* error covariance after calibrated reads; the existing E03M witness is a constructed mathematical example only. E09's optional Ising investigation remains distinct; stable nonvolatile MTJs are not assumed to be stochastic p-bits. No F1–F7 method has passed all novelty and physical gates.

As of 4 November 2026: no freshly extracted target-array readout or new 256-by-256 physical array result is claimed.

References

[1] Bonan Zhang, Peter Deaville, and Naveen Verma. Statistical Computing Framework and Demonstration for In-memory Computing Systems. https://www.princeton.edu/~nverma/VermaLabSite/Publications/2022/ZhangDeavilleVerma_DAC2022.pdf, 2022. Access checked 3 November 2026; method excerpts.

[2] Peter Deaville, Bonan Zhang, Lung-Yen Chen, and Naveen Verma. A Maximally Row-Parallel MRAM In-Memory-Computing Macro Addressing Readout Circuit Sensitivity and Area. https://www.princeton.edu/~nverma/VermaLabSite/Publications/2021/DeavilleZhangChenVerma_ESSCIRC2021.pdf, 2021. Access checked 3 November 2026; method excerpts.

[3] A Fully Row/Column-Parallel MRAM in-Memory Computing Macro With Memory-Resistance Boosting and Weighted Multi-Column ADC Readout. https://ieeexplore.ieee.org/document/10801203/, 2025. Access checked 3 November 2026; abstract.

[4] Peng Dang, Yintao He, and Huawei Li. VARA: A Voltage-Aware ReRAM-Based Accelerator for Energy-Efficient Computing. https://arxiv.org/html/2609.00421v1, 2026. Access checked 3 November 2026; method excerpts.

[5] Weichong Chen et al. DIB-CIM: An Energy-Accuracy Balanced Analog Compute-in-Memory MRAM Using Dynamic-Input-Block Size Arithmetic With Dynamic-Resolution ADC. https://ieeexplore.ieee.org/abstract/document/11672681, 2026. Access checked 3 November 2026; abstract.

[6] Jaebeom Park, Seung Eon Hwang, and Jongsun Park. HINT: A Hybrid SRAM-MRAM Compute-In-Memory with INput-aware Skipping SAR-ADC for Energy Efficient Ternary LLMs. https://pure.korea.ac.kr/en/publications/hint-a-hybrid-sram-mram-compute-in-memory-with-input-aware-skippi/, 2026. Access checked 3 November 2026; abstract.

[7] A 492.8-TOPS/W STT-MRAM Sparsity-Adaptive Compute-in-Memory Macro for Edge AI Inference. https://xplorestaging.ieee.org/document/11554148/, 2026. Access checked 3 November 2026; abstract.

[8] Choongseok Song, Yonguk Sim, and Doo Seok Jeong. Input Sparsity-Aware Computing-In-Memory with Bidirectional Conversion-Skippable Analog-to-Digital Converter. https://advanced.onlinelibrary.wiley.com/doi/epdf/10.1002/aisy.202501336, 2026. Access checked 3 November 2026; abstract and method excerpts.

[9] Choongseok Song, Jeeson Kim, and Doo Seok Jeong. Optimal Weight-Splitting in Resistive Random Access Memory-Based Computing-in-Memory Macros. https://advanced.onlinelibrary.wiley.com/doi/full/10.1002/aisy.202200289, 2023. Access checked 3 November 2026; method excerpts.

[10] Brian Crafton, Samuel Spetalnick, Jong-Hyeok Yoon, and Arijit Raychowdhury. Statistical Optimization of Compute In-Memory Performance Under Device Variation. https://bpb-us-e1.wpmucdn.com/sites.gatech.edu/dist/e/220/files/2021/08/Crafton_ISLPED_2021.pdf, 2021. DOI: 10.1109/ISLPED52811.2021.9502484; access checked 4 November 2026; Sections II–III method excerpts.

[11] Emmanouil Arapidis et al. Analysis and Mitigation of IR Drop in Memristor-based AI Hardware Accelerators. https://research.tudelft.nl/en/publications/analysis-and-mitigation-of-ir-drop-in-memristor-based-ai-hardware/, 2026. Access checked 3 November 2026; abstract and indexed excerpts.

[12] Chowdhury, Zand, and Salehi. Dyn-A-Map: Dynamic AI-Model Mapping for robust and accurate MRAM-based analog edge computing. https://www.nature.com/articles/s44335-026-00072-6, 2026. Access checked 3 November 2026; method excerpts.

[13] Saion K. Roy, Han-Mo Ou, Mostafa G. Ahmed, Peter Deaville, Bonan Zhang, Naveen Verma, Pavan K. Hanumolu, and Naresh R. Shanbhag. Compute SNDR-Boosted 22-nm MRAM-Based In-Memory Computing Macro Using Statistical Error Compensation. https://experts.illinois.edu/en/publications/compute-sndr-boosted-22-nm-mram-based-in-memory-computing-macro-u/, 2025. Access checked 3 November 2026; abstract.

[14] CN122738551A: Memory computing integrated chip. https://eureka.patsnap.com/patent/CN122738551A, n.d. Access checked 3 November 2026; reproduced disclosure.

[15] Rung-Bin Lin and Chi-Ming Tsai. Weight-Based Bus-Invert Coding for Low-Power Applications. https://cecs.uci.edu/~papers/compendium94-03/papers/2002/aspdac02/pdffiles/01b_2.pdf, 2002. Access checked 3 November 2026; indexed primary excerpts.

[16] Charbel Sakr, Yongjune Kim, and Naresh Shanbhag. Analytical Guarantees on Numerical Precision of Deep Neural Networks. https://proceedings.mlr.press/v70/sakr17a.html, 2017. Access checked 3 November 2026; abstract and method excerpts.

[17] Mohammad Hasan Ahmadilivani, Jaan Raik, Masoud Daneshtalab, and Maksim Jenihhin. DeepVigor+: Scalable and Accurate Semi-Analytical Fault Resilience Analysis for Deep Neural Networks. https://arxiv.org/html/2410.15742v1, 2024. Access checked 3 November 2026; method excerpts.

[18] David Boetius, Stefan Leue, and Tobias Sutter. Solving Probabilistic Verification Problems of Neural Networks using Branch and Bound. https://proceedings.mlr.press/v267/boetius25a.html, 2025. Access checked 3 November 2026; abstract and method excerpts.

[19] Soojung Bae, Tanner Andrulis, Vivienne Sze, and Joel S. Emer. Small-Noise is All You Need: Fast and Precise Estimation of Compute-in-Memory Accuracy Loss. https://tanner-andrulis.github.io/posts/2026_small_noise/, 2026. Access checked 3 November 2026; public project only.

[20] Sujan Kumar Gonugondla and Naresh R. Shanbhag. IMPQ: Reduced Complexity Neural Networks via Granular Precision Assignment. https://experts.illinois.edu/en/publications/impq-reduced-complexity-neural-networks-via-granular-precision-as/, 2022. Access checked 3 November 2026; abstract.

[21] Tanner Andrulis, Joel S. Emer, and Vivienne Sze. RAELLA: Reforming the Arithmetic for Efficient, Low-Resolution, and Low-Loss Analog PIM: No Retraining Required! https://eems.mit.edu/wp-content/uploads/2023/04/2023_isca_raella.pdf, 2023. Access checked 3 November 2026; method excerpts.

[22] Ali Nezhadi, Odysseas Chatzopoulos, Dimitris Gizopoulos, and Mehdi Tahoori. Trust, but Verify: Reliable Compute-in-Memory via Double-Reference Sensing and Selective Recompute. https://publikationen.bibliothek.kit.edu/1000196610, 2026. Access checked 3 November 2026; abstract.

[23] Pratham Sharma, Mohd Faisal Khan, and Santosh Kumar Vishvakarma. SLICE-CiM: Selective Layer-wise Inference Control Engine with Confidence-Based Early Exit for SRAM based Digital Compute-in-Memory. https://arxiv.org/html/2610.37601v1, 2026. Access checked 3 November 2026; method excerpts.

[24] Sujan K. Gonugondla, Charbel Sakr, Hassan Dbouk, and Naresh R. Shanbhag. Fundamental Limits on the Precision of In-memory Architectures. https://shanbhag.ece.illinois.edu/publications/IMC-Precision-ICCAD-2020.pdf, 2020. Access checked 3 November 2026; method excerpts.

[25] Tanner Andrulis, Ruicong Chen, Hae-Seung Lee, Joel S. Emer, and Vivienne Sze. Modeling Analog-Digital-Converter Energy and Area for Compute-In-Memory Accelerator Design. https://arxiv.org/html/2404.06553v1, 2024. Access checked 3 November 2026; method excerpts.

[26] Humiao Li et al. A lossless and fully parallel spintronic compute-in-memory macro for artificial intelligence chips. https://www.nature.com/articles/s41928-025-01479-y, 2025. Access checked 3 November 2026; abstract.

[27] Mohammad Hizzani, Ming-Jay Yang, Arne Heittmann, George Higgins Hutchinson, Dmitri Strukov, and John Paul Strachan. Impact of memory device precision in analog combinatorial optimization solvers. https://www.nature.com/articles/s44335-026-00082-4, 2026. Access checked 4 November 2026; primary abstract and indexed passages; direct full text blocked.

[28] Qiuyuan Wang et al. Hardware Acceleration of Stochastic Neural Networks Enabled by Bit-Cell Level Co-Design of Magnetic Tunnel Junctions. https://advanced.onlinelibrary.wiley.com/doi/10.1002/advs.78111, 2026. Access checked 3 November 2026; method excerpts prior access.

[29] Seungki Hong, Kyeongwon Jeong, and Taekwang Jang. Snowball: A Scalable All-to-All Ising Machine with Dual-Mode Markov Chain Monte Carlo Spin Selection and Asynchronous Spin Updates for Fast Combinatorial Optimization. https://arxiv.org/html/2601.21058v1, 2026. Access checked 4 November 2026; Section IV-B.2 method excerpts.

[30] Deborah Volpe, Eleonora Raimondo, Andrea Grimaldi, et al. A Scalable and Resource-Efficient Pipelined p-Computer for Probabilistic Ising Machines. https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.70521, 2026. Access checked 4 November 2026; Sections 3.1–3.2 method excerpts.

[31] Yang et al. Sparse ReRAM Engine: Joint Exploration of Activation and Weight Sparsity in Compressed Neural Networks. https://tdr.lib.ntu.edu.tw/handle/123456789/74357?mode=full, 2019. Access checked 3 November 2026; abstract and author retrospective.

[32] Jiang et al. Mixtral of Experts. https://arxiv.org/abs/2401.04088, 2024. Access checked 3 November 2026; abstract.

[33] Yifan Qin et al. When Small Variations Become Big Failures: Reliability Challenges in Compute-in-Memory Neural Accelerators. https://arxiv.org/html/2603.03491v1, 2026. Access checked 3 November 2026; method excerpts.